

Title

Nursing-supported digital intervention for sodium management in progressive chronic kidney disease: feasibility and effectiveness

Abstract

Background: Digital health tools combined with clinical support may enhance lifestyle adherence in chronic kidney disease management. **Objective:** To assess feasibility, engagement, and outcomes of a nurse-coordinated app-based platform for sodium reduction in patients with stages 3-4 chronic kidney disease. **Methods:** Ninety-nine patients (mean age 58 years; 54% male) were randomly assigned to enhanced digital coaching or standard care. The intervention comprised 12 weeks of app-based reminders, educational resources, and weekly nurse consultations. Primary feasibility endpoint was protocol completion rate; secondary endpoints included urinary sodium levels, systolic blood pressure, and dietary knowledge. **Results:** Eighty-five percent of intervention participants completed 8 or more sessions. Mean sodium excretion declined by 18% in the intervention group versus 3% in controls. Blood pressure reduction favored intervention by 6 mm Hg systolic. Secondary outcome of dietary sodium knowledge improved significantly in both groups ($P < 0.05$), with a larger difference favoring intervention ($P = 0.008$). **Conclusion:** The nurse-supported digital platform demonstrated feasibility and favorable trends in sodium reduction and knowledge improvement, though long-term behavioral sustainability requires further investigation.